

Yanze Wu

Cyber Security Engineering Department | **George Mason University**

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EDUCATION

- **George Mason University** Virginia, U.S.
Ph.D. in Information Technology. GPA 3.95 *Jan. 2024 – Present.*
- **Hohai University** Changzhou, China
Master of Communication and Information Systems. *Sep. 2020 – Jun. 2023*
- **Hohai University** Changzhou, China
Bachelor of Electronic Science and Technology. *Sep. 2016 – Jun. 2020*

AWARDS

- **Graduate Student Travel Fund** Virginia, U.S.
Awarded to students giving presentation and provides financial support for conference-related travel. *Sept. 2025.*
- **Daniel R. Bannister PhD fellowship** Virginia, U.S.
Supported by the Bannister family. Awarded to the top 4 graduate students at GMU. *Dec. 2024.*
- **National Scholarship** Changzhou, China
National Level. Awarded to top 2% students in the college. *Oct. 2022.*
- **First-class Scholarship** Changzhou, China
Bachelor's and Master's Level. Awarded to top 10% students in the college. *2017/2018/2019/2020/2021.*

SKILLS AND INTERESTS

- **Programming skills:** Verilog, C, Python (have experience building hardware accelerators for cryptographic algorithms, modular multiplications, and neural networks.) **Board:** ZYNQ, VCK190, Artix-7
- **Research interests:** Hardware accelerating, spatial computing architecture, post-quantum cryptography, side-channel attack, neural network, ising model.

PUBLICATIONS

- Andrew Fan, **Y. Wu**, Tanvir Arafin. “GPU Acceleration of the Sum-Check Protocol Over Binary Tower Fields for Verifiable Computing ”, DATE’25, Verona, Italy.
- **Y. Wu**, Qian Wang, Tanvir Arafin. “SHAFI: Securing Hash-Based Post-Quantum Cryptography from Hardware Fault Injection Attacks”, AsianHOST’25, Nanjing, China.
- **Y. Wu**, Tanvir Arafin. “Energy-Efficient Acceleration of Hash-Based Post-Quantum Cryptographic Schemes on Embedded Spatial Architectures”, PACT’25, Irvine, California, USA. **(27.8% acceptance rate, 34 of 122)**

- **Y. Wu**, Tanvir Arafin. “Accelerating Torus Fully Homomorphic Encryption on Energy-Efficient Heterogeneous Architecture” , ICCD’25, Dallas, USA. **(25.5% acceptance rate)**
- **Y. Wu** and M. T. Arafin, “ArKANE: Accelerating Kolmogorov-Arnold Networks on Reconfigurable Spatial Architectures,” IEEE Embedded Systems Letters, 2025 **(Impact Factor 1.7)**.
- **Y. Wu** and M. T. Arafin, “Ising Model Processors on a Spatial Computing Architecture,” 2024 IEEE 67th International Midwest Symposium on Circuits and Systems (MWSCAS), Springfield, MA, USA, 2024, pp. 1383-1387, doi: 10.1109/MWSCAS60917.2024.10658926.
- Y. Cao, **Y. Wu**, L. Qin, S. Chen, and CH. Chang, “Areas, Time and Energy Efficient Multicore Hardware Accelerator for Extended Merkle Signature Scheme” in IEEE TCAS-1: Regular Paper, Early Access, 2022, doi: 10.1109/TCSI.2022.3200987. **(Impact Factor 1.18)**
- Y. Cao, **Y. Wu**, X. Lu, S. Chen, J. Ye and CH. Chang, “An Efficient Full Hardware Implementation of Extended Merkle Signature Scheme” in IEEE TCAS-1: Regular Papers, vol.69, no.2, pp. 682-693, Feb. 2021. **(Impact Factor 1.18)**
- W. Zheng, **Y. Wu**, et al. “Accelerating hybrid and compact neural networks targeting perception and control domains with coarse-grained dataflow reconfiguration” 2020, J. Semiconductor., vol. 41, no. 2. **(Impact Factor 0.23)**

TALKS AND PRESENTATIONS

- - **Adaptive Acceleration Arrays for Cryptography:** Achieve high-performance cryptographic algorithm on spatial computing architecture. *George Mason University, Apr. 2024*
 - **Security on Heterogeneous Architecture:** Discussing the transition of hybrid computing architecture and security issues exposed on modern multiple-processor systems. *George Mason University, Feb. 2024*

EXPERIENCE

- **Fisilink Microelectronics Technology Co., Ltd.** Suzhou, China
Internship of Digital Integrated Circuit Design *Jun. 2020 - Sep. 2020*
 - **XMSS Hardware Accelerators:** Learn and implement XMSS in hardware.
 - **Electronic Control Unit:** Construct fast verify modules for vehicles at the standard of 2200 times per second.
 - **Fast Modular Application Accelerators:** Construct fast modular accelerators for high-performance NTT implementation.
- **University of Chinese Academy of Sciences** Shenzhen, China
Research Assistant *Oct. 2019 - Jan. 2020*
 - **Implementation of LSTM Hardware Accelerators:** Learn Long Short-Term Memory (LSTM) neural networks and construct hardware accelerators.
 - **Accelerators for Float-point Numbers:** Build lightweight accelerators for float-point numbers utilized in neural networks.

REFERENCE

- **Tanvir Arafin, Ph.D.** marafin@gmu.edu
Assistant Professor, Cyber Security Engineering Department, George Mason University